Optimization and Curve Sketching

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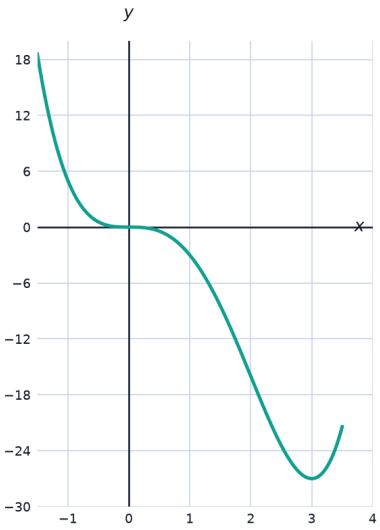
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Critical points and the first/second derivative tests

Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

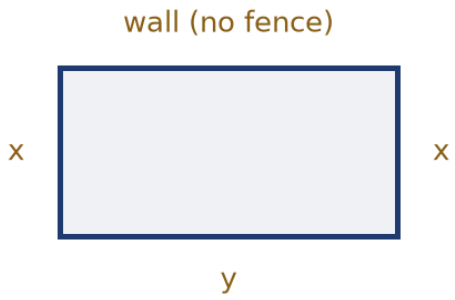
2 [Foundational] • Target: Critical points and the first/second derivative tests

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.



3 [Foundational] • Target: Optimization applications

A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.



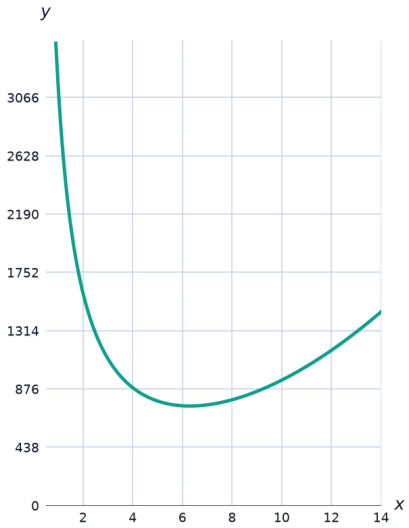
4 [Foundational] • Target: Optimization applications

An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

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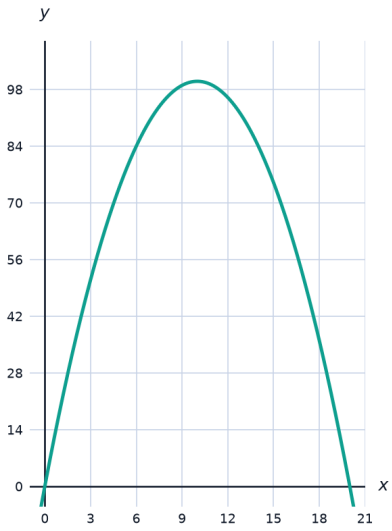
5 [Foundational] • Target: More optimization

A cylindrical can must have volume 500π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).



6 [Foundational] • Target: More optimization

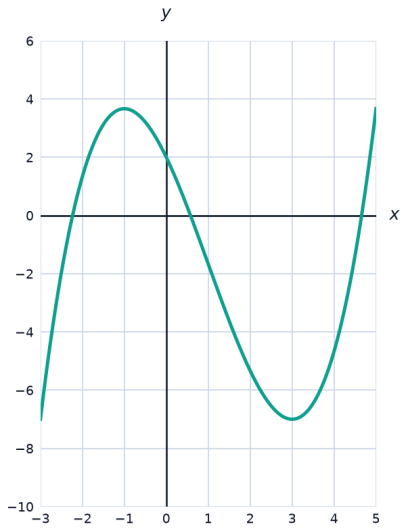
Find two positive numbers whose sum is 20 and whose product is as large as possible.



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7 [Foundational] • Target: Curve sketching

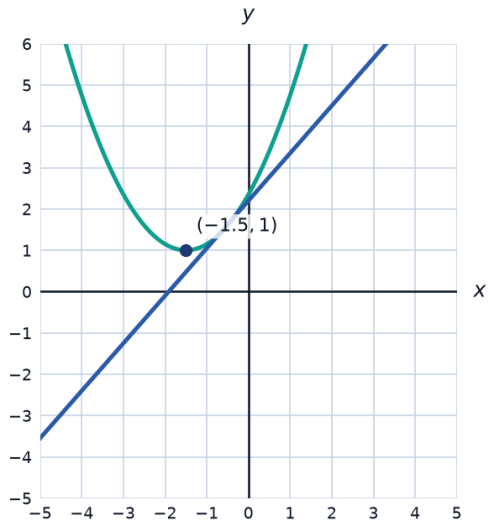
For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Curve sketching

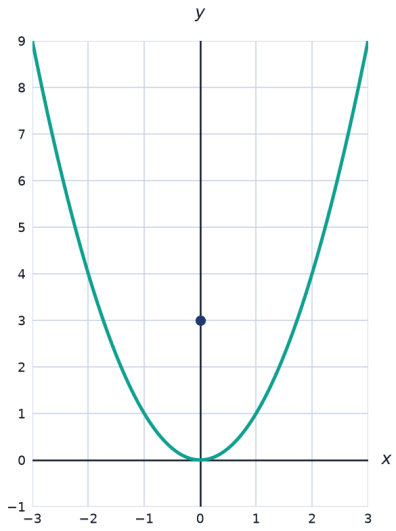
Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.



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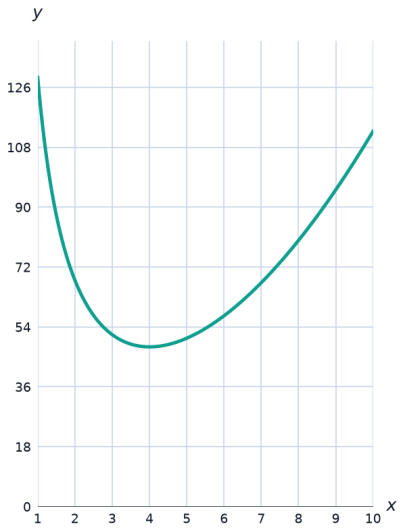
9 [Intermediate] • Target: Minimizing distance and cost

Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.



10 [Intermediate] • Target: Minimizing distance and cost

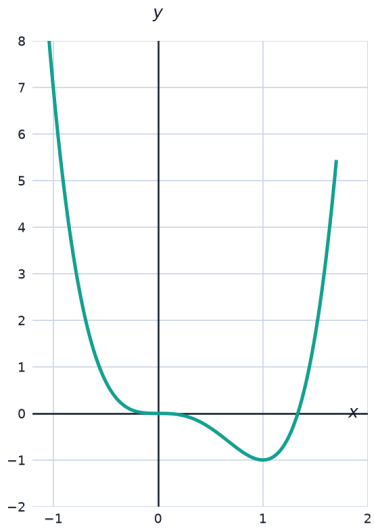
A rectangular box with a square base and open top must have volume $32\ \text{m}^3$. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2 h = 32$).



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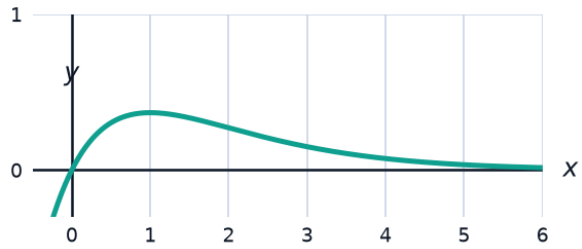
11 [Intermediate] • Target: Sketching using derivative information

For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.



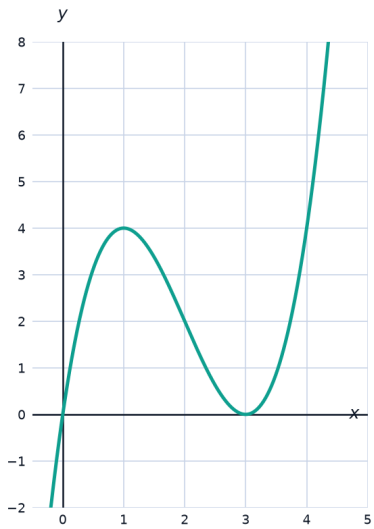
12 [Intermediate] • Target: Sketching using derivative information

For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.



13 [Intermediate] • Target: Visualizing critical points on a graph

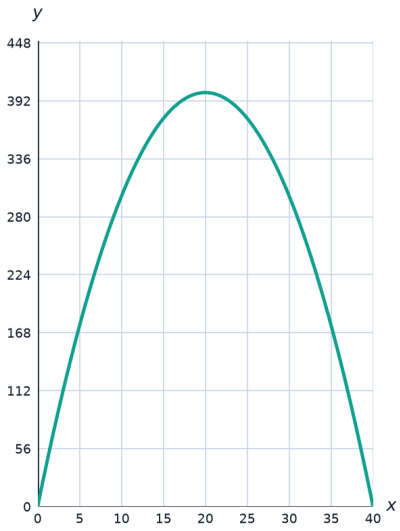
The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.



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14 **[Intermediate]** • **Target: Optimization with a fixed perimeter**

A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.



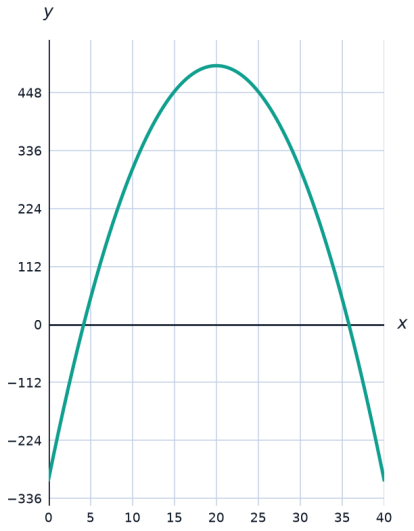
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Optimization with a fixed perimeter** • **Cross-Lesson: 16.01 Vector Fields and Line Integrals**

A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y: $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).

16 **[Challenging]** • **Target: Concavity and the second derivative test in context** • **Cross-Lesson: 2.01 Evaluating Limits**

A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.



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17 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 2.02 Continuity

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \le 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.

18 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 3.01 The Derivative from First Principles

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Critical points and the first/second derivative tests • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Optimization applications • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical points and the first/second derivative tests

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:

a. inconclusive

b. proves local min

c. proves local max

d. proves inflection

22 [Foundational] • Target: Optimization applications

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:

a. local max

b. local min

c. inflection point

d. saddle point

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23 [Foundational] • Target: More optimization

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.

a. $local minimum at x = -\frac{2}{3}, local maximum at x = \frac{2}{3}$

b. $local maximum at x = -\frac{2}{3}, local minimum at x = \frac{2}{3}$

c. both are local minima

d. both are local maxima

24 [Foundational] • Target: Curve sketching

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. inconclusive

25 [Foundational] • Target: Minimizing distance and cost

Critical points occur where $f'(x)$ is:

a. positive

b. negative

c. zero or undefined

d. equal to $f(x)$

26 [Foundational] • Target: Sketching using derivative information

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. local min

b. local max

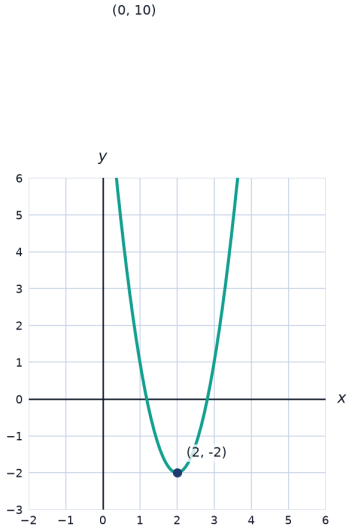
c. inflection

d. saddle

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27 [Foundational] • Target: Visualizing critical points on a graph

Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.



a. 15

b. 9

c. 11

d. 21

28 [Foundational] • Target: Optimization with a fixed perimeter

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_{\alpha}^{\beta} r\,d\theta$

b. $\int_{\alpha}^{\beta} |dr/d\theta|\,d\theta$

c. $\int_{\alpha}^{\beta} r^2\,d\theta$

d. $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Concavity and the second derivative test in context

For $f(x) = x^4 - 8x^2$, inflection points at $x =$

a. 0 only

b. ± 2

c. $\pm\sqrt{\frac{4}{3}}$

d. $\pm 2\sqrt{3}$

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30 [Intermediate] • Target: Critical points and the first/second derivative tests

The second derivative of $f(x) = \sin x$ is:

a. $\cos x$

b. $-\cos x$

c. $-\sin x$

d. $\sin x$

31 [Intermediate] • Target: Optimization applications

Second derivative of $f = \sin(x)$:

a. $-\sin x$

b. $\cos x$

c. $-\cos x$

d. $\sin x$

32 [Intermediate] • Target: More optimization

Find the minimum distance from origin to $y = x^2 + 1$:

a. 1

b. $\sqrt{3/4}$

c. $\sqrt{3}/2$

d. $\sqrt{2}$

33 [Intermediate] • Target: Curve sketching

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$

d. $\pi e^{4\pi}$

34 [Intermediate] • Target: Minimizing distance and cost

To find candidates for the global max/min of f on $[a, b]$, you check critical points and:

a. the endpoints a, b

b. inflection points only

c. only $x = 0$

d. points where f is negative

35 [Intermediate] • Target: Sketching using derivative information

Critical point $x = c$ of f with $f'(c) = 0$: the number of sign changes of f' around c determines:

a. only whether it is max

b. only whether it is min

c. whether it is max, min, or neither

d. the concavity

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36 [Intermediate] • Target: Visualizing critical points on a graph

Second derivative d^2y/dx^2 parametrically:

a. $\frac{d}{dt}(dy/dx) \cdot \frac{dt}{dx}$

b. $(d^2y/dt^2)/(d^2x/dt^2)$

c. d^2y/dt^2

d. dy/dx

37 [Intermediate] • Target: Optimization with a fixed perimeter

Find the inflection points of $f(x) = xe^{-x}$:

a. $x = 1$ only

b. $x = 2$ only

c. $x = 0$ only

d. $x = 0$ and $x = 2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 3.01 The Derivative from First Principles

Acceleration is the second derivative of position. If $x(t) = t^3 - 6t^2 + 9t$, when is the particle at rest?

a. $t = 1, 3$

b. $t = 2$

c. $t = 0, 3$

d. $t = 1, 2$

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39 [Challenging] • Target: Critical points and the first/second derivative tests • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. is the rate of change of the optimal value w.r.t. the constraint constant

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

40 [Challenging] • Target: Optimization applications • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Distance particle travels (vs displacement): use:

a. $\int_a^b v(t)dt$

b. $\int_a^b |v(t)|dt$

c. $v(b) - v(a)$

d. $s(b) - s(a)$

41 [Challenging] • Target: More optimization • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Find the total distance (not displacement) of $v(t) = \cos(\pi t)$ on $[0, 2]$:

a. 0

b. $2/\pi$

c. 1

d. $4/\pi$

42 [Challenging] • Target: Curve sketching • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find all critical points of $f(x) = x^3 - 6x^2 + 9x + 1$.

a. $x = 1$ only

b. $x = 3$ only

c. $x = 1, 3$

d. $x = 0, 1, 3$

43 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 6.01 Related Rates

Analytic function: uniqueness theorem — if f, g analytic on connected open set, and $f = g$ on a sequence of points converging to an interior point, then $f = g$:

a. on the boundary only

b. at all integers

c. at a dense subset

d. everywhere on the domain

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44 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 7.01 Integration by Substitution

$f(x) = x^4 - 4x^3$. Concavity: concave up for:

a. $x > 0$

b. $x > 3$ and $x < 0$

c. $x < 0$

d. $x > 2$

45 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 7.02 Integration by Parts

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

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